AGENTIC AI @ UIUC

Voice Agents.

The next frontier of agentic AI by Agentic AI@UIUC

A G E N D A

01 How we got here

02 The market

03 Why it matters

04 What it actually is

05 Live demo

06 What's next

07 Questions

Anthropic Admitted Claude Code Broke. We Were Right.

Hidden reasoning. A usage reset. Pricing changes.

How

did we get here?

Press 1 for billing.

Phone trees. Rigid menus. No understanding.

2011

“Hey Siri.”

Voice as a command line. One question, one answer.

2022 — 2023

LLMs learn to talk.

Reasoning meets conversation. Still turn-by-turn.

Real-time.
Interruptible.
Human.

Why

does this matter?

3×

Speaking is faster than typing.

It’s how humans already talk to humans.

75%

of customer calls can be handled by voice AI.

The work is already there. Waiting.

$12B

Projected voice AI market by 2028.

Capital is following the conversation.

0 3

What

are we actually building?

A voice agent
listens. understands. reasons. speaks.

In real time. Like a person would.

STT → LLM → TTS

Speech → Text

Reason

Text → Speech

Three stages. One conversation.

300ms

is human.

Anything over a second feels broken.

PIPELINE

STT → LLM → TTS

3 calls. Latency stacks.

REALTIME

Audio → Model → Audio

One model. End to end.

Hears tone. Hears pacing. Hears you cut in.

End-to-end. You ship in a day.

Most popular. Phone + web. Strong tooling.

Retell

Clean UX. Visual workflow builder.

Bland

Phone calls at scale. Enterprise.

INFRASTRUCTURE

Pick your pieces. Wire them yourself.

Deepgram

STT + TTS

Eleven Labs

TTS

OpenAI

LLM + Realtime

LiveKit

Transport

Support

Health

Sales

Learning

01 **Use a platform.** 30 min

02 **Wire the Realtime API.** 2–4 hrs

03 **Build the full pipeline.** 1–2 days

Live Demo.

Triage. Schedule. Confirm.

Indistinguishable from human.

In two to three years.